

Tour de Maths

Leg 2: 27 February 2025

St Johns College

The following questions are each worth 5 points.

Question 1

Each child of the Smith family has at least three sisters and at least one brother.

What is the minimum number of children in this family?

Question 2

A three digit number is divisible by 8, 12 and 130.

What is the smallest such number?

Question 3

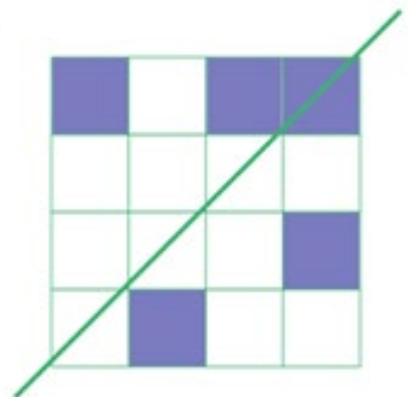
Determine the sum of the digits of the product of $5^{104} \times 4^{52}$.

Question 4

What is the smallest number which must be added to 2010 to arrive at a perfect square?

Question 5

Some squares in a grid have been shaded. If the picture must be symmetrical about the diagonal line, more squares need to be shaded. Once this has been done, what is the maximum number of squares NOT shaded?



Question 6

A jug was 60% full of water. After 20% of that water is removed the jug contains 192ml.

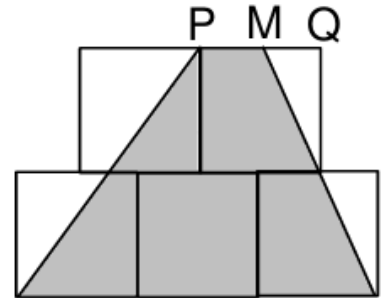
What is the maximum amount of water (in ml) that the jug could contain?

Question 7

Five equal squares each with side 2 cm are used to make the figure alongside.

M is the midpoint of PQ.

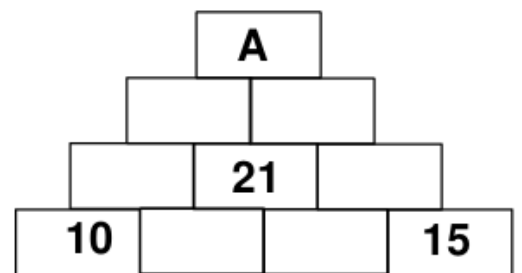
What is the area (in cm^2) of the shaded region?



Question 8

In the given diagram, the number in any box is equal to the sum of the numbers in the two boxes immediately below it.

What is the value of A?



Question 9

A bathroom floor is covered by square tiles: the floor is 5 tiles wide and 8 tiles long. If one of the floor tiles is chosen at random, what is the probability that it is at the edge of the floor?

Question 10

This map shows a grid of one way streets. How many different routes are there from A to B?



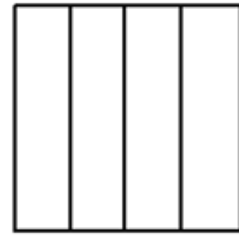
The following questions are each worth 10 points.

Question 11

A square is divided into four identical rectangles as shown.

The perimeter of each square of the four rectangles is 20 units.

What is the perimeter of the square?



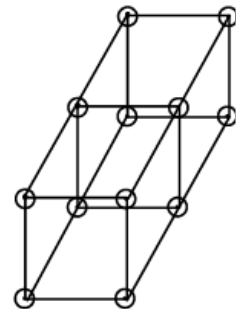
Question 12

A cricket ball and a soccer ball roll in a straight line along the ground and cover the same distance. The cricket ball rotates 27 times, while the soccer ball rotates only 9 times. If the radius of the cricket ball is 4 *cm*, then what is the radius (in *cm*) of the soccer ball?

Question 13

The diagram shows part of some scaffolding which consists of many equal rods joined together by connectors. The diagram shows a structure of length 2 units.

Determine the number of rods required to make a structure of length 40 units.



Question 14

A soccer ball is made up of 12 pentagons (5-sided figures) and 20 hexagons (6-sided figures) which are stitched together along their edges to form seams.

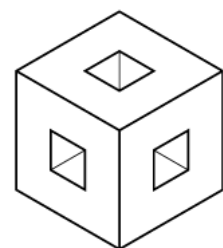
How many seams does the soccer ball have?

Question 15

The mass of a solid $3 \times 3 \times 3$ cube is 810*g*.

Three shafts, each $1 \times 1 \times 3$, are made as shown in the sketch.

What is the mass of the remaining solid?



The following questions are each worth 20 points.

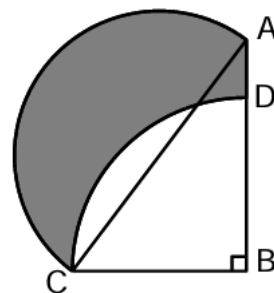
Question 16

At my school, the science club has 15 members and the chess club has 12 members. If 13 members belong only to one of the two clubs, how many learners belong to both clubs?

Question 17

ABC is a right-angled triangle with $BC = 6$ and $AB = 8$. A semicircle with AC as diameter and a quarter circle with BC as radius are drawn.

What is the area of the shaded region, rounded to the nearest integer?



Question 18

Anne says Barbara is lying.

Barbara says Catherine is lying.

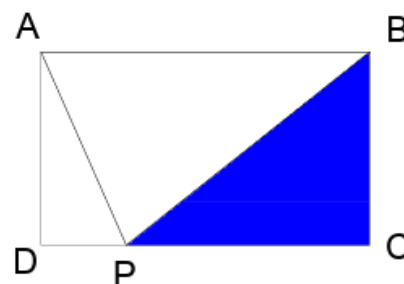
Catherine says Barbara is lying.

Diane says Anne is lying.

How many girls are lying?

Question 19

The area of $\triangle ADP$ is $\frac{1}{5}$ the area of the rectangle. What is the proportion of the rectangle that is shaded?



Question 20

In the diagram, $AB = AC$, $BC = BD$ and $DE = AD$.

If $\angle EDB = 30^\circ$, then determine the size of $\angle BAC$.

